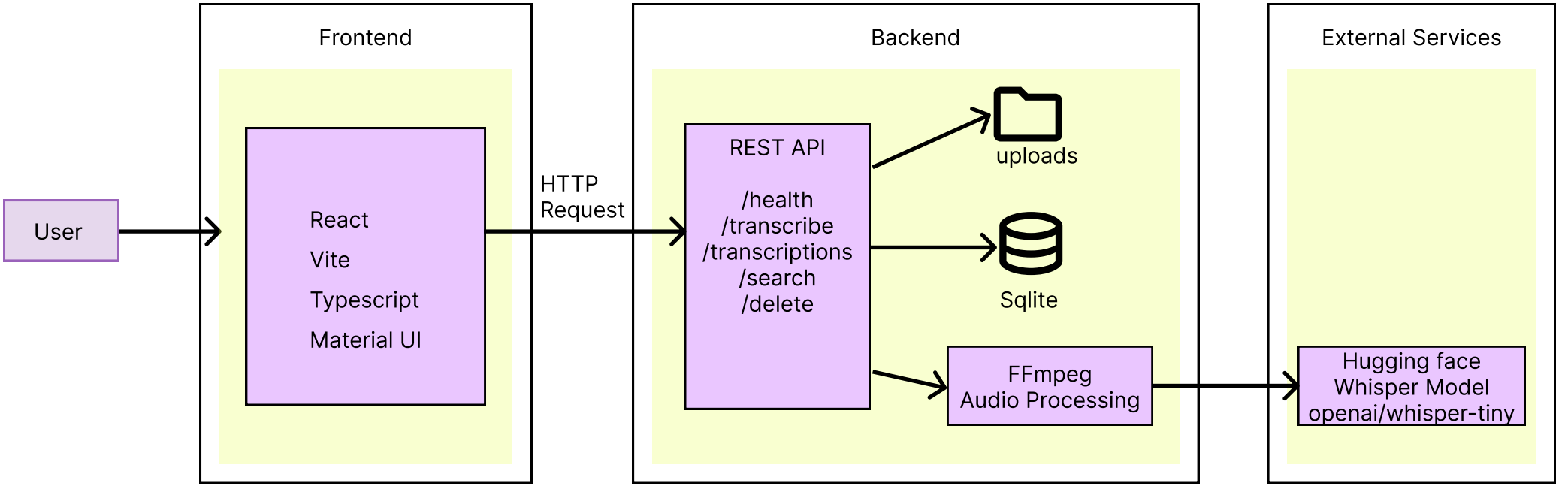

1. The frontend uses react + vite + typescript application to handle the interface for uploading audio files, view, search and delete transcriptions.
2. The frontend communicates with the backend over HTTP using REST endpoints.
3. The backend receives request, validates, business logic and return responses.
4. The audio files are uploaded into file storage in the uploads folder. This storage is separate from the database because binary files are being handled as files on disk rather than stored as blobs in database.
5. The transcriptions are stored in the sqlite database, with metadata such as filename, filepath, transcript and timestamp in transcriptions.db.
6. FFmeg handles media preprocessing is used to decode uploaded audio files to process audio and passed to the whisper model.
7. The whisper model transform audio into transcript and hand it back to the API layer for storage and response.